

1.are hereditary units transmitted from generations to generations

- A. Gametes
- B. Zygotes
- C. Genes**
- D. Sex cells

2. According to the theory of pangenesis, every inherited structure contributed a little amount of itself called to the gonads

- A. Blood vessels
- B. Genetics
- C. Gemmules**
- D. Gametes

3. The sex cells in humans are made in the.....

- A. Gonads**
- B. Gonadotropin
- C. Ovules
- D. Scrotum

4. The theory of pangenesis was proposed by

- A. August Weissman
- B. Charles Darwin**
- C. Aristotle
- D. Karl Ernst Von Beer

5. The theory that suggests that acquired features can be passed on to offspring is the theory of

- A. Pangenesis**
- B. Preformation
- C. Epigenesis
- D. Lamarckism

6. The germplasm theory suggests that theare responsible for the transfer of heritable traits

- A. Somatic cells
- B. Germ cells**
- C. Egg cells
- D. Sperm cells

7. According to the theory of preformation, the preformed man was referred to as.....

A. **Homunculus**

B. Humanculas

C. Homicide

D. Humus

8. The theory of preformation was proposed by

A. Charles Darwin

B. Carl Ernst Von Beer

C. **Jan Swammerdam and Charles Bonnet**

D. Jean Baptiste Lamarck

9. Which theory suggests the sex cells are homogenous bits of organic matter?

A. The theory of Lamarckism

B. The theory of preformation

C. The germplasm theory

D. **The theory of Epigenesis**

10. Which theory suggests that development is differentiation as well as growth?

A. The theory of Lamarckism

B. The theory of preformation

C. The Germplasm theory

D. **The theory of Epigenesis**

11. The different forms of a gene that determine alternative traits is referred to as

A. Chromosome

B. **Alleles**

C. Chromatids

D. Chromatin

12. Dominancy is only experienced in

A. Homozygous

B. **Heterozygous**

C. Homologues

D. Chromosomes

13. Offsprings of the cross of P1 individuals are referred to as

A. F2 generation

B. P generation

C. **F1 generation**

D. F3 generation

14 .A trait that escapes the F1 generation but it is expressed in the F2 generation may be a

- A. **Recessive**
- B. Dominant
- C. Co-dominant
- D. Autosomal

15._____refers to a heterozygous hybrid concerning a specified gene

- A. Monohybrid
- B. **Dihybrid**
- C. Trihybrid
- D. Monoploid

16._____refers to the visible expression of a trait or it is any measurable characteristic or distinctive trait possessed by an organism

- A. **Phenotype**
- B. Genotype
- C. Trait
- D. Homologues

17. Recessive alleles are expressed only in____ Individuals.

- A. **Homozygous**
- B. Heterozygous
- C. Dominant
- D. Homologous

18. Human somatic cells contain
_____chromosomes

- A. **46**
- B. 23
- C. 46 pairs
- D. 22 pairs

19. There are____pairs of autosomes and _____ pair of sex chromosomes in humans

- A. **22, 1**
- B. 23,2
- C. 44,2
- D. 2, 46

20.In meiosis I, a diploid mother cell produces__daughter cells

- A. **2 haploid**
- B. 4 haploid

- C. 2 diploid
- D. 4 monohybrid

21. Meiosis I and II produce ____ daughter cells

- A. 2 haploid
- B. **4 haploid**
- C. 2 diploid
- D. 4 diploid

22. Chromosome replication occurs at the ____ phase of interphase

- A. G1
- B. G2
- C. **S**
- D. G0

23. The anaphase I of meiosis I differs from mitotic anaphase in that in meiotic anaphase I

- A. Centromere divides
- B. **Sister chromatids divide**
- C. Centromere does not divide
- D. Tetrad does not separate

24. Mitosis can be arrested at the late prophase by a chemical called

- A. Colchicine
- B. **Colchicine**
- C. Chloroquine
- D. Vinblastin

25. The prophase of mitosis is characterized by

- A. Formation of nuclei
- B. Alignment of chromosomes at the equator
- C. **Formation of spindle fibres**
- D. Division of cytoplasm

26. Meiosis occurs in

- A. Somatic cells
- B. **Sex cells**
- C. Nerve cells
- D. Body cells

27. Mitosis occurs in

- A. **Somatic cells**

- B. Sex cells
- C. Haploid cells
- D. Egg cells only

28. One characteristic of the mitotic telophase is the

- A. Reformation of spindle fibres
- B. Alignment of chromosomes at the equator
- C. **Reformation of nuclei**
- D. Chromosome duplication

29. During prophase I of meiosis I, the homologous chromosomes pair to form

- A. **Tetrad**
- B. Zygote
- C. Embryo
- D. Chromatin

30. The first meiotic division is said to be a ____ division

- A. Homozygous
- B. Heterozygous
- C. Holandric
- D. **Heterotypic**

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1. Recessive Alleles are identified experimentally by

- A. By observing their dominance in crosses
- B. By crossing potential carriers with homozygous dominant individuals**
- C. By crossing potential carriers with heterozygous individuals
- D. By examining their influence on multiple traits

2. What term describes zygotes carrying two identical alleles for a gene?

- A. Heterozygous
- B. Dominant
- C. Homozygous**
- D. Recessive

3. What is the term used to refer to the full set of chromosomes from an individual?

- A. Genome
- B. Karyotype**
- C. Chromosome set
- D. Gene map

4. Which of the following syndromes is associated with an additional copy of chromosome 21?

- A. Edwards syndrome
- B. Patau syndrome
- C. Down syndrome**
- D. Warkany syndrome 2

5. Which condition results from a trisomy of chromosome 18?

- A. Down syndrome
- B. Patau syndrome
- C. Edwards syndrome**
- D. Turner syndrome

6. Which type of chromosomal abnormality involves gaining two extra chromosomes?

- A. Trisomy
- B. Tetrasomy**
- C. Pentasomy
- D. Nullisomy

7. Which of the following describes Pentasomy?

- A. Loss of one chromosome
- B. Gain of one extra chromosome
- C. Gain of two extra chromosomes**
- D. Gain of three extra chromosomes

8. What is the result of a complete loss of a pair of homologous chromosomes?

- A. Trisomy
- B. Tetrasomy
- C. Pentasomy
- D. Nullisomy**

9. Which term refers to an organism with a single set of chromosomes?

- A. Diploid
- B. Polyploid
- C. Monoploid**
- D. Euploid

10. Which of the following is true about X-linked recessive disorders?

- A. Affected females cannot pass the disorder to their sons and daughters
- B. Affected males can pass the disorder to their sons
- C. More females are affected than males
- D. More males are affected than females**

11. A newborn is diagnosed with Klinefelter syndrome, which results from the presence of an extra X chromosome (47, XXY). Which of the following symptoms might this child experience later in life?

- A. Overproduction of testosterone
- B. Development of female secondary sexual characteristics**
- C. Increased muscle mass
- D. High fertility

12. A chromosomal analysis shows a translocation between chromosomes 9 and 22, often referred to as the "Philadelphia chromosome." This translocation is associated with which of the following conditions?

- A. Down syndrome
- B. Turner syndrome
- C. Chronic myeloid leukemia**
- D. Huntington's disease

13. A genetic disorder is caused by a single extra chromosome in the affected individuals' cells. During which of the following stages might this error most likely occur?

- A. Prophase of mitosis
- B. Anaphase of mitosis
- C. Anaphase I of meiosis**
- D. Telophase II of meiosis

14. A couple with no history of cystic fibrosis has a child diagnosed with the disease. What can be inferred about the inheritance pattern and genotype of the parents?

- A. Cystic fibrosis is dominant, and both parents are homozygous recessive.
- B. Cystic fibrosis is recessive, and both parents are carriers (heterozygous).**
- C. The disease is sex-linked, and the father is affected.
- D. The disease is dominant, and one parent is affected.

15. In pea plants, the allele for purple flowers (P) is dominant over the allele for white flowers (p). If a heterozygous purple-flowered plant is crossed with a white-flowered plant, what is the probability of obtaining a white-flowered offspring?

- A. 0%
- B. 25%
- C. 50%**
- D. 75%

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16. A genetic test reveals a deletion on chromosome 5, known as "cri du chat" syndrome, characterized by a high-pitched cry in infants. What is the most likely outcome of this chromosomal deletion?

- A. Normal development and lifespan
- B. Increased risk of heart disease only
- ☒ C. Delayed physical and cognitive development
- D. Enhanced muscular strength

17. An amniocentesis shows an abnormal karyotype with a Robertsonian translocation between chromosomes 13 and 14. What is a possible implication of this finding in future offspring?

- A. Increased risk of cystic fibrosis
- B. Higher likelihood of Turner syndrome
- ☒ C. Risk of having a child with trisomy 13
- D. High probability of unaffected offspring only

18. During cell division, a scientist notices that the chromosome number in a cell is halved, resulting in four non-identical cells. Which process is most likely being observed?

- A. DNA replication
- ☒ B. Meiosis I
- C. Meiosis II
- D. Mitosis

19. A mutation in a gene that regulates the cell cycle results in uncontrolled cell division. Which phase of the cell cycle is likely impacted by this mutation?

- A. Cytokinesis
- ☒ B. G1 phase
- C. G2 phase
- D. S phase

20. In humans, brown eye color (B) is dominant over blue eye color (b). A woman with brown eyes (Bb) and a man with blue eyes (bb) have a child. What is the probability that the child will have blue eyes?

- A. 0%
- B. 25%
- ☒ C. 50%
- D. 100%

21. A child is diagnosed with trisomy 18, also known as Edwards syndrome, characterized by severe developmental delays and physical abnormalities. Which of the following is a likely cause of this chromosomal condition?

- A. Crossing over between non-homologous chromosomes
- B. Deletion on chromosome 18
- C. Duplication of chromosome 18 during mitosis
- ☒ D. Nondisjunction during meiosis

22. An adult male with a 47, XYY karyotype is often taller than average and may have learning difficulties. Which of the following statements best describes this chromosomal condition?

- A. A duplication of the X chromosome causes it.
- B. It arises from a deletion on the Y chromosome.
- ☒ C. It is due to nondisjunction during spermatogenesis
- D. It results from the fusion of two Y chromosomes.

23. A researcher observes that a cell in metaphase has all its chromosomes aligned at the cell's equator. Which phase will directly follow metaphase, and what key event will occur?

- ☒ A. Anaphase; sister chromatids separate to opposite poles
- B. Cytokinesis; the cell divides into two identical cells
- C. Prophase; chromosomes condense further
- D. Telophase; the nuclear envelope disintegrates

24. If a woman who is a carrier for hemophilia (an X-linked recessive disorder) has children with a man without hemophilia, what is the probability that their son will have hemophilia?

- A. 0%
- B. 25%
- ☒ C. 50%
- D. 100%

25. In a population, the allele for black fur (B) in rabbits is dominant over white fur (b). A homozygous black rabbit is crossed with a homozygous white rabbit. What fur color will the offspring likely have?

- A. 50% black and 50% white fur
- B. 75% black and 25% white fur
- ☒ C. All will have black fur
- D. All will have white fur

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- C. Chromosome se
- D. Gene map

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- ☒ B. Tetrasomy
- C. Pentasomy
- D. Nullisomy

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30. Which of the following describes pentasomy?

- A. Gain of one extra chromosome
- ☒ B. Gain of three extra chromosomes
- C. Gain of two extra chromosomes
- D. Loss of one chromosome

42. Which term refers to an organism with a single set of chromosomes?

- A. Diploid
- B. Euploid
- ☒ C. Monoploid
- D. Polyploid

31. Which chromosomal rearrangement involves the duplication of a chromosomal segment?

- A. Deletion
- ☒ B. Duplication
- C. Inversion
- D. Translocation

32. Which condition is caused by duplication of the gene encoding peripheral myelin protein 22 (PMP22) on chromosome 17?

- ☒ A. Charcot-Marie-Tooth disease type 1A
- B. Down syndrome
- C. Edwards syndrome
- D. Patau syndrome

33. What type of chromosomal rearrangement involves the exchange of segments between two different chromosomes?

- A. Duplication
- B. Inversion
- ☒ C. Reciprocal translocation
- D. Ring Formation

34. Which chromosomal abnormality only occurs with chromosomes 13, 14, 15, 21, and 22?

- A. Inversion
- B. Isochromosome
- C. Ring formation
- ☒ D. Robertsonian translocation

35. Which chromosomal abnormality occurs when a chromosome segment is broken, inverted, and reattached?

- A. Deletion
- B. Duplication
- ☒ C. Inversion
- D. Translocation

36. Which chromosomal rearrangement involves a portion of one chromosome being deleted from its normal place and inserted into another chromosome?

- A. Duplication
- B. Insertion
- C. Inversion
- ☒ D. Translocation

37. What is a diagram showing the ancestral relationships and transmission of genetic traits over several generations in a family called?

- A. Family tree
- B. Gene map
- C. Karyotype

☒ D. Pedigree

38. During which phase of mitosis does the nuclear envelope break down?

- A. Anaphase
- B. Metaphase
- ☒ C. Prophase
- D. Telophase 1

39. If the sex of an individual is unknown in a pedigree, what shape is used?

- A. Circle
- ☒ B. Diamond
- C. Square
- D. Triangle
- A. M
- B. S

40. During which stage of prophase I in meiosis do chromosomes condense and appear as threads?

- A. Diplonema (Diplotene)
- ☒ B. Leptonema (Leptotene)
- C. Pachynema (Pachytene)
- D. Zygonema (Zygotene)